

Hand-in assignment 2

Instructions

- This assignment consists of **two questions**.
- On Vula it opens at 12:00 on Friday, 22 March 2024, please upload your solutions under **Vula Assignments no later than 23:55 on Tuesday, 02 April 2024**. Make sure that no pages or questions are missing.
- Write your full name and student number on your solutions.
- If possible, **upload a single document** saved as **ABCXYZ001_A2** (where **ABCXYZ001** is your campus Id).
- Show all your work or reasoning. **Answers without work or reasoning shown where needed will receive no mark.**

Questions

1. State whether the following are true or false. If true, prove it or if false, give a counter-example.
 - a) Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function. If X is finite, then f is continuous. [3]
 - b) Let (X, d_X) and (Y, d_Y) be metric spaces. If $f : X \rightarrow Y$ is a continuous bijection, then f^{-1} is continuous. [3]
 - c) Let (X, d_X) and (Y, d_Y) be metric spaces, then every isometry $f : X \rightarrow Y$ is surjective. [3]
 - d) Let (X, d_X) and (Y, d_Y) be metric spaces, then every Lipschitz-continuous function $f : X \rightarrow Y$ is injective. [3]
 - e) Let (X, d_X) and (Y, d_Y) be metric spaces, then every Lipschitz-continuous function $f : X \rightarrow Y$ is uniformly continuous. [3]
2. List all possible topologies on a two element set. Are there different topologies on such a set such that they result in homeomorphic topological spaces? If any, make the homeomorphism explicit. [7]